

Homework 7: Robust Regression

Submission Notes: Please write your answers to each question under that question (not altogether at the end). It will be easier for me to give full credit if your answer is commented well, including comments about where you are completing each sub-part of the question. For an example, see Homework 1.

1. Below, I simulate data with heteroskedastic error, noting the variance is a function of the predictor x .

```
set.seed(1933)
x <- 1:10
(beta0 <- runif(1, -5, 5))

## [1] 0.08868237

(beta1 <- runif(1, -5, 5))

## [1] -2.116944

y <- beta0 + beta1 * x + rnorm(10, mean = 0, sd = x) # Variance is  $x^2$ 
dat <- tibble(x = x,
              y = y)
```

When I fit an OLS model, it doesn't "pick up" the regression equation.

```
mod.ols <- lm(y~x, data=dat)
summary(mod.ols)$coefficients

##               Estimate Std. Error   t value   Pr(>|t|)
## (Intercept) -6.4290175  2.8929497 -2.222305 0.05697684
## x           -0.7669293  0.4662411 -1.644920 0.13860636
```

In Figure 1, we see that the increased variance almost creates a quadratic (if we ignore the observation at $x = 10$)

```
ggplot(data=dat)+
  geom_point(aes(x=x, y=y))+
  theme_bw() +
  geom_abline(aes(intercept=coef(mod.ols)[1],
                  slope=coef(mod.ols)[2],
                  color="OLS"))+
  geom_abline(aes(intercept=beta0,
                  slope=beta1,
                  color="Truth"))+
  labs(color="")
```

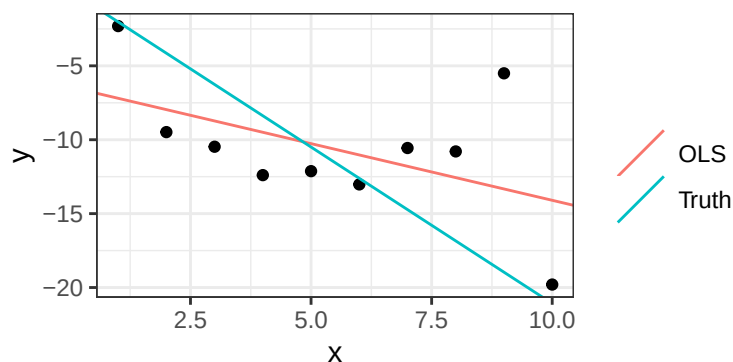


Figure 1: Heteroskedastic regression data generated for Question 1.

- (a) Conduct a residual and influence analysis on the `mod.ols`.

Note: The *known* issue may not be obvious.

```
source("plotResiduals.R")
source("plotInfluence.R")

plotResiduals(mod.ols)
```

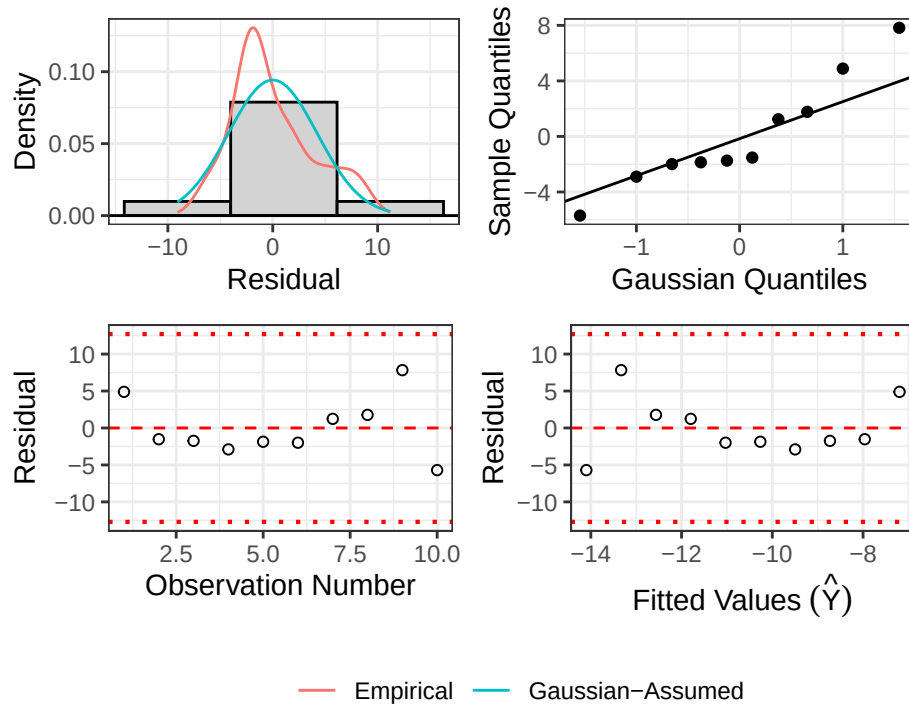


Figure 2: Residual Analysis on `mod.ols`

If I was analyzing the residuals of this model I would assume there was a non-linear relationship between the predictors and the response variable due to the clear curved pattern in the residuals. The histogram of the residuals looks a little skewed but it's not terrible for 10 observations. The qqplot indicates a potential non-linear relationship with the quadratic looking pattern on the right tail.

```
plotInfluence(mod.ols)
```

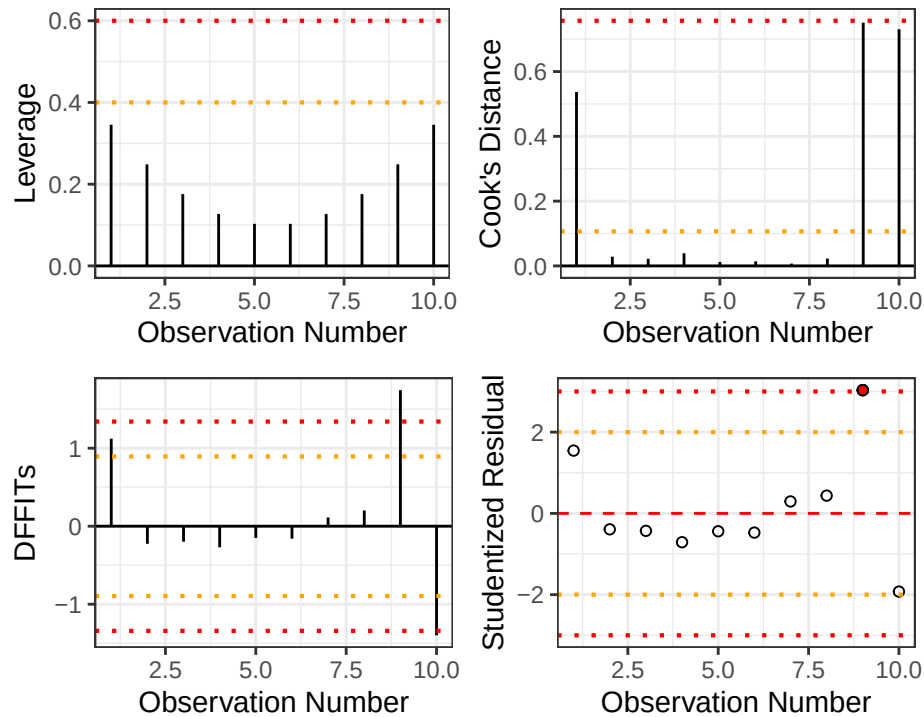


Figure 3: Influence Analysis on `mod.ols`

Looking at the influence, the thing that stands out is the clear leverage pattern across observation numbers, which would indicate to me that the observations are not independent of each other. The studentized residuals show similar problems to the residuals in Figure: 1a. Additionally, with there only being 10 observations, several of them having large Cook's d and DFFITS is a bit concerning due to the sheer proportion of potentially high influence points making up a large part of the model.

(b) Fit a weighted least squares (WLS) regression model.

i. Why is $w_i = 1/x_i^2$ a sensible choice for the weights?

Hint: Why is the RMSE ≈ 1 ?

In the setup of the regression equation the variance at each point $x(1:10)$ is x^2 , so weighting each point by $\frac{1}{x^2}$ should make each point have weighted variance 1.

ii. Fit the model with these weights, how does the result change?

```
weights = 1/x^2

mod_weighted <- lm(y ~ x, data = dat, weights = weights)

(sum <- summary(mod_weighted))

##
## Call:
## lm(formula = y ~ x, data = dat, weights = weights)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.0210  -0.6560   0.0217   0.7014   1.4473
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.7035      1.3762  -1.238  0.25088
## x             -1.8698      0.5418  -3.451  0.00868 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.145 on 8 degrees of freedom
## Multiple R-squared:  0.5982, Adjusted R-squared:  0.548
## F-statistic: 11.91 on 1 and 8 DF, p-value: 0.008678
```

Now that we used the weighted model, the coefficient β_1 associated with x is estimated to be -1.8698 ($p = 0.00868$). This is only a few tenths greater than the true value of $\beta_1 = -2.1169$. The intercept β_0 is estimated to be -1.7035 while the true β_0 is 0.0887 , indicating that the estimate for the intercept is still quite off. This would probably be better if the data extended more for $x \leq 0$.

iii. Add the WLS line to the scatterplot.

```
betas <- sum$coefficients
beta0_w <- betas[1,1]
beta1_w <- betas[2,1]

ggplot(data=dat)+
  geom_point(aes(x=x, y=y))+
  theme_bw() +
  geom_abline(aes(intercept=coef(mod.ols)[1],
                  slope=coef(mod.ols)[2],
                  color="OLS"))+
  geom_abline(aes(intercept=beta0,
                  slope=beta1,
                  color="Truth"))+
  geom_abline(aes(intercept = beta0_w,
                  slope=beta1_w,
                  color="WLS")) +
  labs(color="") +
  geom_hline(yintercept = 0, linetype="dashed")
```

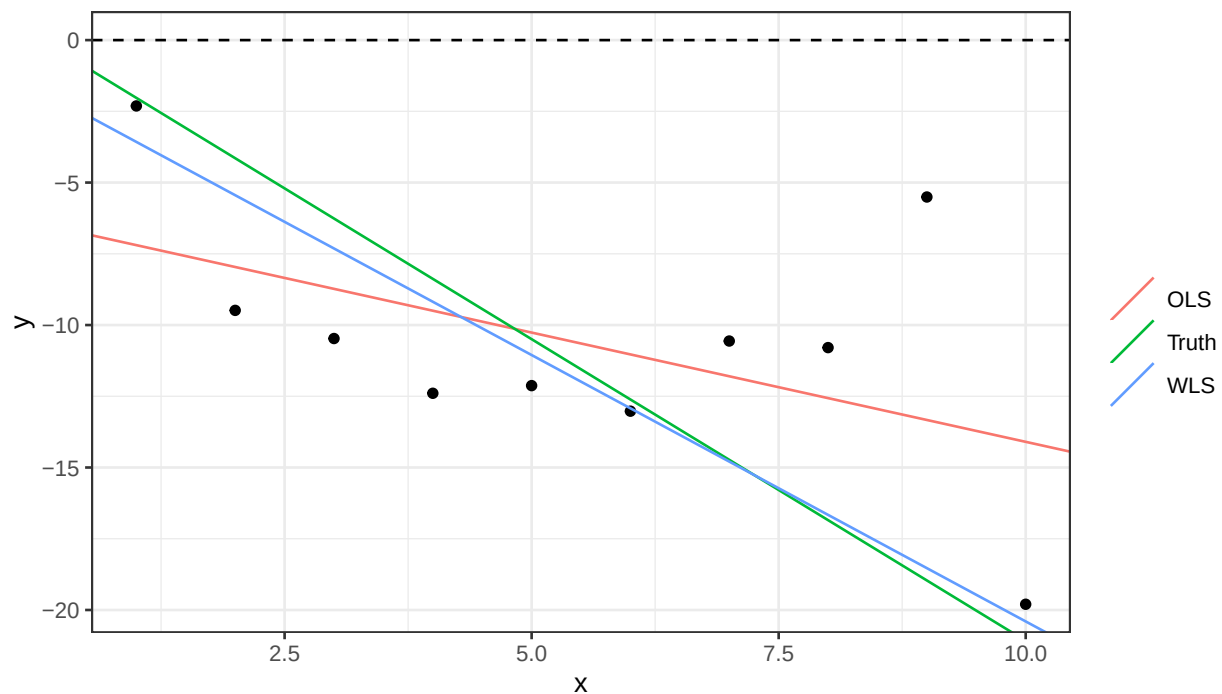


Figure 4: Scatterplot with Estimated Regression Lines (Updated With Weighted OLS Model)

The weighted model worked as intended here, as clearly the WLS regression line is much closer to the truth than the OLS regression line.

2. Explore fitting a regression model to the data simulated below.

```

n <- 50
set.seed(7272)
x <- sample(x=seq(0,100,0.01), size=n, replace=TRUE)
e <- rnorm(n=50, mean=0, sd=5)
y <- 3.5 + 2.1*x + e

first_dat <- tibble(x = x,
                    y = y)

# Add a misbehaved point
x2 <- c(x, 100)
y2 <- c(y, 25)

```

- (a) Fit an OLS model based on x and y . Check the assumptions of the model and interpret the coefficients of the model; are they what you would expect?

```
first_mod <- lm(y ~ x, data = first_dat)
```

```

first_mod <- lm(y ~ x, data = first_dat)
plotResiduals(first_mod)

```

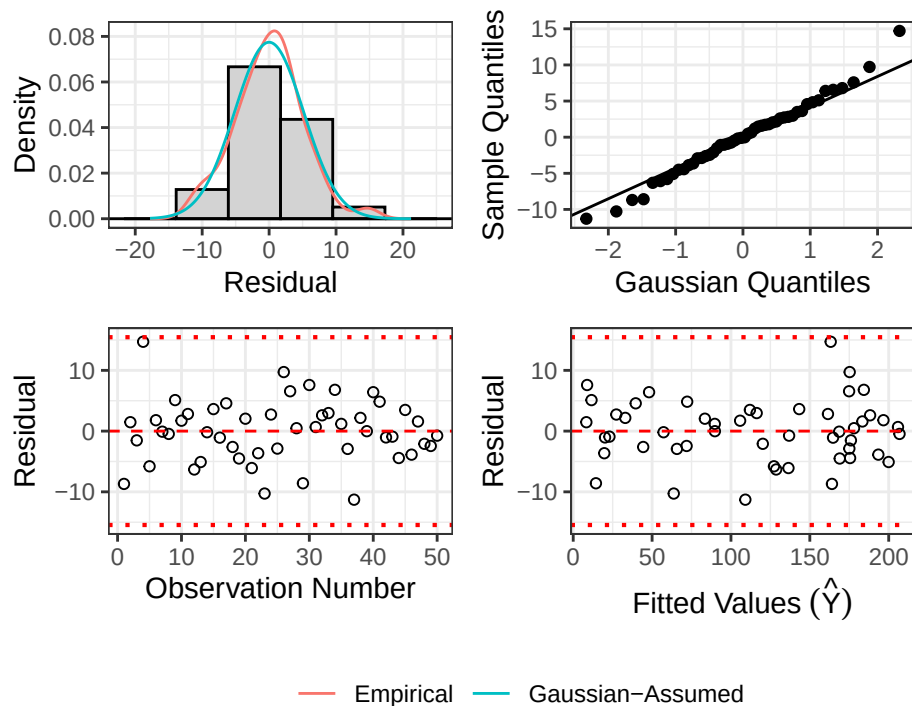


Figure 5: Residual Analysis on x and y OLS model

The residuals look perfect. They're centered at 0 with good normality. The qqplot is very reasonable. Finally, the residuals are indiscriminantly scattered with constant variance.

```
plotInfluence(first_mod)
```

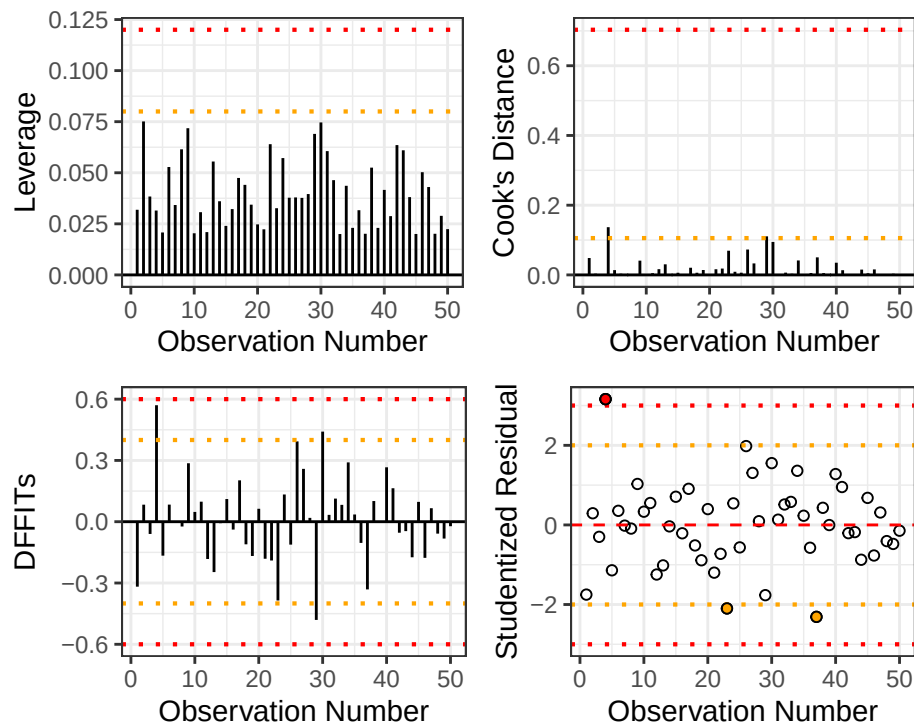


Figure 6: Influence Analysis on x and y OLS model

- (b) Fit an OLS model based on x_2 and y_2 . Check the assumptions of the model and interpret the coefficients of the model; are they what you would expect?
3. Continue with Question 2.
 - (a) Fit an IRWLS model based on x_2 and y_2 using Huber weights. Interpret the coefficients of the model; are they what you would expect?
 - (b) Fit an IRWLS model based on x_2 and y_2 using Bisquare weights. Interpret the coefficients of the model; are they what you would expect?
 - (c) Fit an IRWLS model based on x_2 and y_2 using `lmrob()` from the `robustbase` package for R. Interpret the coefficients of the model; are they what you would expect?
 - (d) Fit a Quantile regression model based on x_2 and y_2 . Interpret the coefficients of the model; are they what you would expect?
4. **Optional, but recommended** Plot the results! Create a scatterplot of x_2 and y_2 . Superimpose the true regression line, the OLS estimate, the IRWLS estimates, and the quantile regression estimate. Interpret the results.
5. **Optional, but recommended** Complete Questions 3-4, but use $n = 1000$. What changes? Why do you think that is?